Free-Body Diagrams and Net Force — teacher key

Question: Which forces act on an object, and which way do they add up?

Outline and pacing (42 min)

Tab	Priority	Minutes	If behind
start	1	2	never trimmed
warmup	1	6	never trimmed
predict	1	3	never trimmed
explore	3	3	skipped first
models	2	8	shrinks to 3 of 5
build	2	8	shrinks to 6 of 12
practice	1	5	never trimmed
check	1	3	never trimmed
Huddle h1	clock	minute 16	pulled in by the clock

Targets

- I can name the two objects in any force interaction.
- I can sort forces as contact or field forces.
- I can draw a free-body diagram and find the net force.

Predictions

- A book rests on a table. How many forces act on the book? **Two: gravity down and the table's push up.** Gravity pulls down and the table pushes up. They are equal, so the net force is zero.
- A crate slides to the right at constant speed. A person pushes it. How does the push compare with friction? **The push equals friction.** Constant speed means zero net force, so the push and friction are equal.
- The same crate now speeds up to the right. Which way does the net force point? **To the right.** The net force points the way the crate speeds up: the push beats friction.

Huddle script (answers in green)

Definitions

- **Free-body diagram:** A **picture** of one object with an **arrow** for each force on it.
- **Normal force:** The push of a **surface** on an object, **perpendicular** to the surface.
- **Net force:** The **sum** of all the forces on an object, with **direction**.

Worked example

A crate has W = 40 N down, N = 40 N up, a push of 20 N right, and friction of 8 N left. Find the net force.

- Up and down: 40 N and 40 N **cancel**.

- Right and left: 20 − 8 = **12** N.
- Net force: 12 N to the **right**.
- The crate **speeds up** to the right.

Label the crate

1. The **applied force** pushes right.
2. **Friction** opposes the slide.
3. **Gravity** pulls down.
4. The **normal force** pushes up.

Every level, with answers

Name the objects in an interaction

Level 1. A person pushes a wall. Which object exerts the push on the wall?
(A) The person (B) The wall (C) The floor (D) The air
Answer: A The person's hands touch the wall and push it.
Misconception targeted: Students let an object exert a force on itself, or name "motion" as the source.

Level 2. A book rests on a table. Which two objects interact through the normal force?
(A) The book and the table (B) The book and the floor (C) The book and the air (D) The table and the floor
Answer: A The normal force is a push between two surfaces that touch: the book and the table.
Misconception targeted: Students name one object, or a pair that does not touch.

Level 3. Which force would still act on a satellite-free rock in empty space near the Earth?
(A) Gravity (B) Normal force (C) Friction (D) Tension
Answer: A Gravity is a field force: the Earth pulls without touching. Normal force, tension and friction need contact.
Misconception targeted: Students call gravity a contact force because it "touches the ground".

Level 4. A banner hangs from a rope fixed to the ceiling. Which list names every object that exerts a force ON the banner?
(A) The rope and the Earth (B) The rope, the Earth and the ceiling (C) The rope only (D) The ceiling and the Earth
Answer: A The rope touches the banner and pulls up. The Earth pulls down at a distance. The ceiling touches the rope, not the banner.
Misconception targeted: Students include every object in the scene, not just those acting on this object.

Sort and name the forces

Level 1. A rough floor opposes a sliding box. This force is called...
(A) Friction (B) Normal force (C) Gravity (D) Tension
Answer: A A force that opposes sliding between two surfaces is friction.
Misconception targeted: Students mix up normal force and gravity because both are vertical.

Level 2. A book rests on a flat table. Which way does the normal force on the book point?
(A) Up (B) Down (C) Left (D) Right
Answer: A The normal force is perpendicular to the surface and pushes away from it: up.

Misconception targeted: Students think friction points the way the object moves, or that gravity stops at the top of a throw.

Level 3. Is applied force a contact force or a field force?
(A) Contact force (B) Field force (C) Neither
Answer: A Applied force needs the two objects to touch: a contact force.
Misconception targeted: Students call gravity a contact force, or call normal force a field force.

Level 4. A cart is pushed right across a rough floor. Which list names every force on it?
(A) W, N, F, f (B) W, N, F (C) W, N, F, f, force of motion (D) F, f
Answer: A Gravity (W) and the floor's normal force (N) act on it, plus the pull or push and friction from the rough floor. Nothing else touches it.
Misconception targeted: Students add a "force of motion", or forget the normal force and friction.

Pick the free-body diagram

Level 1. A phone lies on a shelf. Which diagram shows the forces on the phone?
(A) W down, N up; N equals W (B) W down (C) W down, N up, F right; N equals W (D) W down, N up; N shorter than W
Answer: A Gravity pulls down and the surface pulls or pushes up with an equal arrow, so the net force is zero.
Misconception targeted: Students draw only gravity on a still object, or add a force for "staying there".

Level 2. A crate slides left at constant speed. A person pushes it left. Which diagram is right?
(A) W down, N up, F left, f right; N equals W; F equals f (B) W down, N up, F left; N equals W (C) W down, N up, F left, f right, M left; N equals W; F equals f (D) W down, N up, F left, f right; N equals W; F longer than f
Answer: A Constant speed means net force zero: push equals friction, and N equals W.
Misconception targeted: Students believe motion needs a bigger forward force than friction, or draw a "force of motion".

Level 3. A trunk speeds up to the right. A person pushes it right. Which diagram is right?
(A) W down, N up, F right, f left; N equals W; F longer than f (B) W down, N up, F right, f left; N equals W; F equals f (C) W down, N up, F right, f left, M right; N equals W; F longer than f (D) W down, N up, F right; N equals W
Answer: A Speeding up means the net force points right: the push arrow is longer than friction. N still equals W.
Misconception targeted: Students draw push and friction equal for any motion.

Level 4. A person pushes a crate right with 35 N. Friction is 10 N and the crate weighs 40 N. It speeds up. Which diagram has arrows drawn to scale?
(A) W down (40 N), N up (40 N), F right (35 N), f left (10 N); N equals W; F longer than f (B) W down (40 N), N up (30 N), F right (35 N), f left (10 N); N shorter than W; F longer than f (C) W down (40 N), N up (40 N), F right (10 N), f left (35 N); N equals W; F shorter than f (D) W down (40 N), N up (40 N), F right (35 N); N equals W
Answer: A N equals W (40 N). The right arrow (35 N) is longer than the left (10 N), so the net force is 25 N to the right.
Misconception targeted: Students draw lengths that do not match the numbers, or let N differ from W.

Which way is the net force?

Level 1. A box has 6 N pushing right and 6 N pushing left. Which way is the net force?
(A) It is zero (B) Right (C) Left

Answer: A Equal arrows in opposite directions cancel: the net force is zero.
Misconception targeted: Students think equal forces still "push" the box one way.

Level 2. A crate has 9 N pulling left and 7 N of friction pulling right. Which way is the net force?
(A) Left (B) Right (C) It is zero
Answer: A The bigger force is 9 N, left. The net force is 2 N left.
Misconception targeted: Students take the net force along the motion, or call friction the "smaller force" and ignore it.

Level 3. A crate weighs 60 N, the floor pushes up 60 N, a person pushes right 8 N, and friction is 4 N left. Which way is the net force?
(A) Right (B) It is zero (C) Left (D) Up
Answer: A Up and down cancel (60 N each). Right 8 N and left 4 N: net 4 N right.
Misconception targeted: Students try to add vertical and horizontal forces into one number.

Level 4. A 3 kg lamp (weight 29.4 N) hangs from a chain pulling up 18 N. Which way is the net force?
(A) Down (B) Up (C) It is zero
Answer: A $W = 3 \times 9.8 = 29.4$ N. The chain pulls 18 N. The weight is bigger, so the net force points down.
Misconception targeted: Students assume a hanging object is always balanced.

Find the net force

Level 1. A cart is pushed right with 9 N and friction pulls left with 4 N. How big is the net force?
Answer: 5 N $9 - 4 = 5$ N, to the right.
Misconception targeted: Students add forces that point opposite ways.

Level 2. A crate is pulled right with 12.5 N. Friction is 8.5 N left. How big is the net force?
Answer: 4 N $12.5 - 8.5 = 4$ N.
Misconception targeted: Students drop the decimal or add the forces.

Level 3. A 5 kg box (weight 49 N, floor pushes up 49 N) is pushed right with 15 N. Friction is 10 N. What is the net force?
Answer: 5 N Up and down cancel. Net $= 15 - 10 = 5$ N, to the right.
Misconception targeted: Students include weight in a horizontal net force.

Level 4. Two people push a crate right with 23 N and 14 N while friction pulls left with 15 N. What is the net force?
Answer: 22 N Right: $23 + 14 = 37$ N. Left: 15 N. Net $= 37 - 15 = 22$ N right.
Misconception targeted: Students subtract in the order they read the numbers.

Two objects, one diagram

Level 1. A cup sits on a desk. Which force does NOT act ON the cup?
(A) The cup's push on the desk (B) Gravity on the cup (C) The desk's push up on the cup
Answer: A The cup pushes on the desk, so that force acts on the desk. A diagram of the cup shows only forces on the cup.
Misconception targeted: Students draw the object's force on the table as a force on the object.

Level 2. A book A sits on a book B, and B sits on a table. Which list names the forces on book A?
(A) Gravity on A, and B's push up (B) Gravity on A, B's push up, and the table's push up (C) Gravity on A and on B (D) Gravity on A only

Answer: A A is touched only by B, so B pushes up on it. The table touches B, not A.
Misconception targeted: Students list every force in the picture on every object.

Level 3. Box A (4 kg) sits on box B (1 kg). The table pushes up on B. How hard does the table push, in newtons? (g = 9.8 N/kg)
Answer: 49 N The table holds up both: N = (4 + 1) × 9.8 = 49 N.
Misconception targeted: Students count only the lower box's weight.

Level 4. Boxes of 2 kg and 3 kg are tied together and pulled right with 34 N. Friction on the pair totals 12 N. What is the net force on the pair?
Answer: 22 N 34 – 12 = 22 N to the right. The tension between the boxes is inside the system and does not count.
Misconception targeted: Students add the tension between the boxes into the net force.

The six moves

1. **Sketch and name the system.** Draw the situation. Circle the one object you will write the law for.
2. **Draw the free-body diagram.** One arrow per force ON that object. Label each as "force by what, on what". Equal forces, equal lengths.
3. **Choose the positive direction.** Pick the way the object accelerates, or the way it would. Forces that point that way are positive.
4. **Write the sum of forces.** $\Sigma F = ma$ with every term signed. Write it in symbols before any number goes in.
5. **Solve with units.** Isolate the unknown, substitute, carry units. N is $kg \cdot m/s^2$.
6. **Sense-check.** Direction right? Size believable? Does it pass a limit (zero mass, zero push)?

Model problems (five, with every step)

M1 (I do): A 2.0 kg book rests on a table

Find the normal force on the book.
Predict: Before any math: is the table pushing harder than, equal to, or less than the pull of gravity? Why? **Equal. The book is not accelerating, so the arrows cancel.**

1. **Sketch and name the system.** System: the book. Interactions: the book with Earth (field) and the book with the table (contact).
2. **Draw the free-body diagram.** FBD: W down (Earth on book), N up (table on book). Same length.
3. **Choose the positive direction.** Positive: up.
4. **Write the sum of forces.** $\Sigma F = N - W = ma$, and $a = 0$.
5. **Solve with units.** $N = W = mg = (2.0\ kg)(9.8\ m/s^2) = 19.6$ N.
6. **Sense-check.** Check: up, same size as the weight. Pass. Note N equals W because $a = 0$, not because they are a third-law pair.

Answer: 19.6 N *Watch for: Calling N and W an action-reaction pair. They act on the same object, so they cannot be.*

M2 (I do): A 12 kg crate is pushed right with 80 N; friction is 50 N

Find its acceleration.
Predict: Does it move at constant speed, speed up, or slow down? Which way does the net force point? **Speeds up to the right. The push beats friction, so the net force points right.**

1. **Sketch and name the system.** System: the crate.
2. **Draw the free-body diagram.** FBD: F = 80 N right, f = 50 N left, W and N cancel and are equal and opposite.
3. **Choose the positive direction.** Positive: right.
4. **Write the sum of forces.** $\Sigma F = F - f = ma$. Up and down add to zero.
5. **Solve with units.** $a = (80 - 50)\ N / 12\ kg = 2.5\ m/s^2$.
6. **Sense-check.** Check: positive means right, matches the prediction. 30 N on 12 kg is a bit over 2. Pass.

Answer: 2.5 m/s^2 *Watch for: Adding friction to the push. Friction points against the slide, so its sign is negative.*

M3 (we do): A 60 kg student stands on a scale in an elevator that speeds up upward at 1.5 m/s^2

What does the scale read?
Predict: Heavier than, equal to, or lighter than the 588 N at rest? Why? **Heavier. To accelerate upward the floor must push up more than gravity pulls down.**

1. **Sketch and name the system.** System: the student, not the elevator.
2. **Draw the free-body diagram.** FBD: N up (scale on student), W down. N is longer than W.
3. **Choose the positive direction.** Positive: up.
4. **Write the sum of forces.** $\Sigma F = N - mg = ma$.
5. **Solve with units.** $N = m(g + a) = 60(9.8 + 1.5) = 678$ N.
6. **Sense-check.** Check: bigger than 588 N. If $a = 0$ the formula gives mg. Pass.

Answer: 678 N; follow-up 498 N *Watch for: Thinking the scale reads weight. A scale reads the normal force on you.*

M4 (we do): A 4.0 kg block on a frictionless table is tied to a 2.0 kg hanging block over a light pulley

Find the acceleration and the string tension.
Predict: Is the tension more than, equal to, or less than the 19.6 N weight of the hanging block? **Less. The hanging block speeds up downward, so gravity beats the string.**

1. **Sketch and name the system.** Two objects, so two diagrams. Block 1: T right (string), W and N cancel. Block 2: T up, W2 down.
2. **Draw the free-body diagram.** Same string, same T. Same speed, same a.
3. **Choose the positive direction.** Positive: the direction of motion, right for block 1 and down for block 2.
4. **Write the sum of forces.** Block 1: $T = m1\ a$. Block 2: $m2\ g - T = m2\ a$.

5. **Solve with units.** Add: m2 g = (m1 + m2) a, so a = 19.6/6.0 = 3.27 m/s^2. Then T = m1 a = 13.1 N.
6. **Sense-check.** Check: T < 19.6 N. If m1 is huge, a → 0. If m1 → 0, a → g. Pass.

Answer: 3.27 m/s^2 and 13.1 N *Watch for: Setting T equal to the hanging weight. That is only true if the block is not accelerating.*

M5 (you do): Two blocks touch on a frictionless floor: 3.0 kg and 5.0 kg. A 24 N push acts on the 3.0 kg block

Find the force the 3.0 kg block exerts on the 5.0 kg block.
Predict: More than 24 N, equal, or less? Which block feels more net force? **Less than 24 N. The 5.0 kg block gets only part of the push.**

1. **Sketch and name the system.** Draw the system of both blocks first: only the 24 N is external. a = 24/8.0 = 3.0 m/s^2.
2. **Draw the free-body diagram.** Now the 5.0 kg block alone: the contact force C is the only horizontal force.
3. **Choose the positive direction.** Positive: right.
4. **Write the sum of forces.** C = m2 a.
5. **Solve with units.** C = 5.0 × 3.0 = 15 N.
6. **Sense-check.** Check: 15 < 24. By the third law the 5.0 kg block pushes back on the 3.0 kg block with 15 N, leaving it 9 N: 9/3.0 = 3.0. Pass.

Answer: 15 N *Watch for: Saying the 5.0 kg block feels the full 24 N.*

The six hidden bands, one sample each

Students never see a band name. Rung A is bands 1–2, B is 3–4, C is 5–6. Build stops at band 4; bands 5 and 6 appear in Practice and Check.

Band 1 (core, rung A)

A rope pulls a sled. Which two objects interact, and is the force contact or field?
(A) rope and sled, contact (B) rope and sled, field (C) one object alone, contact (D) motion and rope, contact
Answer: A
Every force has two objects: one pushes or pulls, one is pushed or pulled. Touching means contact. Acting across a gap means field.
Misconception targeted: A force belongs to one object; a force never comes from "motion".

Band 2 (honors, rung A)

A rope pulls a 2 kg bucket straight up with 27.6 N. Take up as positive. Find the acceleration.
Answer: 4 m/s^2
W = 19.6 N down. ΣF = T − W = 8 N. a = 4 m/s^2 upward.
Misconception targeted: Forgetting weight when a rope pulls.

Band 3 (AP Physics 1, rung B)

On a frictionless floor, a 20 N push acts on a 4 kg block that touches a 6 kg block. How big is the contact force between them?
Answer: 12 N Symbolic: C = F m2 / (m1 + m2)
System: a = F/(m1 + m2) = 2 m/s^2. Second block alone: C = m2 a = 12 N.
Misconception targeted: Saying the second block feels the full push.

Band 4 (AP C / F=ma, rung B)

Mass 6 kg hangs from mass 1 kg, which hangs from a cable. The elevator holding the cable accelerates up at 1 m/s^2. Find the tension in the cable. Then find the tension between the masses.
Answer: 75.6 N and 64.8 N Symbolic: T1 = (m1 + m2)(g + a), T2 = m2 (g + a)
Whole stack: T1 – (m1 + m2) g = (m1 + m2) a. Lower mass alone: T2 – m2 g = m2 a.
Misconception targeted: Giving the same tension to both strings.

Band 5 (olympiad, rung C)

A uniform 4 kg rope of length 2 m hangs from a 55.2 N upward pull on its top end. Find the tension in the rope 1 m below the top.
Answer: 27.6 N Symbolic: T(s) = F (1 – s/L)
Whole rope: a = (F – Mg)/M = 4 m/s^2 up. The piece below depth s has mass M(1 – s/L): T = M(1 – s/L)(g + a) = F(1 – s/L). The tension falls linearly to zero at the free end.
Misconception targeted: Assuming one tension for the whole rope.

Band 6 (college, rung C)

A uniform chain of length 2.5 m lies on a frictionless table with 1.2 m hanging over the edge. Find its acceleration at that instant. Then show that y(t) = y0 cosh($\sqrt{}$(g/L) t).
Answer: 4.7 m/s^2 Symbolic: a = g y / L
Only the hanging part is pulled: (λ y) g = (λ L) a. a = g y/L, so $\ddot{y}$ = (g/L) y, which has cosh solutions.
Misconception targeted: Using a = g for a chain that is partly on a table.

Common misconceptions

Students say	Say instead
The crate needs a force of motion pushing it forward.	Motion is not a force. Name the object that exerts every arrow. At constant speed the net force is zero.
A book at rest has no forces on it, or only gravity.	The table pushes up. At rest the forces balance: N equals W.
The ceiling pulls on the hanging lamp.	Only objects touching the lamp (the chain) and the Earth act on it.
Gravity is a contact force.	Gravity is a field force. It acts without touching.
Friction points the way the object moves.	Friction opposes the sliding between surfaces.
Add all the numbers to get the net force.	Add forces that point the same way. Subtract opposite ones. Vertical and horizontal do not mix.